

A NEW FORMULA FOR SUMS OF POWERS USING CENTRAL NEWTON'S INTERPOLATION

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ABSTRACT.

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PRELIMINARIES

Lemma 0.1 (Stirling form of Binomial). *For integers n, k*

$$\binom{n}{k} = \sum_{j=0}^k \frac{(-1)^{k-j}}{k!} \begin{bmatrix} k \\ j \end{bmatrix} n^j$$

1. DEFINITIONS AND NOTATIONS

Definition 1.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

Definition 1.2 (Central factorials). *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = n \left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0, \end{cases}$$

where $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=1}^{k-1} \left(n + \frac{k}{2} - j\right)$ are falling factorials.

2. HISTORICAL CONTEXT

3. CENTRAL NEWTON'S FORMULA

Proposition 3.1. (Central Newton's formula [?, p. 462]). *For arbitrary integers x , and t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where $\delta^k f(t)$ denotes central finite difference

$$\delta^k f(t) = \sum_{j=0}^k (-1)^j \binom{k}{j} f\left(t + \frac{k}{2} - j\right).$$

4. HELPER LEMMAS

Lemma 4.1 (Binomial form of central factorials). *For integers n and $k \geq 1$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \binom{n + \frac{k}{2} - 1}{k-1}_{k-1} = \frac{n}{k(k-1)!} \binom{n + \frac{k}{2} - 1}{k-1}_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$. □

Lemma 4.2 (Halved central factorials). *For integers $n, k \geq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \binom{n + \frac{k}{2}}{k} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k}.$$

Proof. We have $\binom{a}{k} = \frac{a}{k} \binom{a-1}{k-1}$. Let $a = n + \frac{k}{2}$, then

$$\binom{n + \frac{k}{2}}{k} = \frac{n + \frac{k}{2}}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Thus,

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \binom{n + \frac{k}{2}}{k} - \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \left[2 \binom{n + \frac{k}{2}}{k} - \binom{n + \frac{k}{2} - 1}{k-1} \right] = \frac{1}{2} \left[\binom{n + \frac{k}{2}}{k} + \binom{n + \frac{k}{2} - 1}{k} + \binom{n + \frac{k}{2} - 1}{k-1} - \binom{n + \frac{k}{2} - 1}{k-1} \right].$$

Thus, the statement follows. □

Proposition 4.3 (Central Newton's formula for powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$n^m = \sum_{k=0}^m \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

Proposition 4.4 (Halved central factorial powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n-t+\frac{k}{2}}{k} + \binom{n-t+\frac{k}{2}-1}{k} \right] \delta^k t^m$$

Proposition 4.5 (Generalized hockey-stick identity). *For integers a, b and j ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right)$. By hockey stick identity $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^b \binom{k}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

Lemma 4.6 (Centered hockey-stick identity). *For integers $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Proof. By setting $j = 1 - t + \frac{k}{2} + r$, and $b = n - t + \frac{k}{2} + r$ yields

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k}.$$

By Generalized hockey-stick identity (4.5),

$$\sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Thus, the claim follows. □

5. ORDINARY SUMS OF POWERS

Proposition 5.1 (Ordinary sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \delta^k t^m \sum_{j=1}^n \left[\binom{j-t+\frac{k}{2}}{k} + \binom{j-t+\frac{k}{2}-1}{k} \right]$$

Proposition 5.2 (Ordinary sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right] \delta^k t^m$$

Proposition 5.3 (Central factorial numbers sums). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n+\frac{k}{2}+1}{k+1} - \binom{1+\frac{k}{2}}{k+1} + \binom{n+\frac{k}{2}}{k+1} - \binom{\frac{k}{2}}{k+1} \right] k! T(m, k),$$

where $T(n, k) = \frac{\delta^{k0} n^m}{k!}$ denote Central factorial numbers of the second kind (in Riordan's notation, see [?, p. 213], and [?].), defined by central Newton's formula for powers, evaluated in $t = 0$

$$n^m = \sum_{k=0}^{\infty} T(m, k) n^{[k]}.$$

Example 5.4 (Sums of squares in central differences).

$$\begin{aligned} t = 0 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(1+n) + \frac{\delta^2 t^2}{2} \frac{n(1+3n+2n^2)}{6}, \\ t = 1 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-1) + \frac{\delta^2 t^2}{2} \frac{n(1-3n+2n^2)}{6}, \\ t = 2 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-3) + \frac{\delta^2 t^2}{2} \frac{n(13-9n+2n^2)}{6}, \\ t = 3 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-5) + \frac{\delta^2 t^2}{2} \frac{n(37-15n+2n^2)}{6}, \\ t = 4 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-7) + \frac{\delta^2 t^2}{2} \frac{n(73-21n+2n^2)}{6}. \end{aligned}$$

Example 5.5 (Sums of cubes in central differences).

$$\begin{aligned}
t = 0 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(1+n) + \frac{\delta^2 t^3}{2} \frac{n(1+3n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-1+n+4n^2+2n^3)}{24}, \\
t = 1 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-1) + \frac{\delta^2 t^3}{2} \frac{n(1-3n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(1+n-4n^2+2n^3)}{24}, \\
t = 2 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-3) + \frac{\delta^2 t^3}{2} \frac{n(13-9n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-21+25n-12n^2+2n^3)}{24}, \\
t = 3 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-5) + \frac{\delta^2 t^3}{2} \frac{n(37-15n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-115+73n-20n^2+2n^3)}{24}, \\
t = 4 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-7) + \frac{\delta^2 t^3}{2} \frac{n(73-21n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-329+145n-28n^2+2n^3)}{24}.
\end{aligned}$$

6. REMARK ABOUT CENTRAL FACTORIAL NUMBERS OF THE SECOND KIND

7. GENERALIZED CENTRAL FACTORIAL NUMBERS OF THE SECOND KIND

Definition 7.1 (Generalized Central factorial numbers of the second kind). *For integers $0 \leq k \leq m$, and an arbitrary integer t ,*

$$T_t(m, k) = \frac{\delta^k t^m}{k!},$$

defined by central Newton's formula for powers, evaluated at arbitrary integer t

$$n^m = \sum_{k=0}^{\infty} T_t(m, k)(n-t)^{[k]}.$$

Proposition 7.2 (Generalized Central factorial sums). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right] k! T_t(m, k)$$

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	0	1							
2	0	0	1						
3	0	$\frac{1}{4}$	0	1					
4	0	0	1	0	1				
5	0	$\frac{1}{16}$	0	$\frac{5}{2}$	0	1			
6	0	0	1	0	5	0	1		
7	0	$\frac{1}{64}$	0	$\frac{91}{16}$	0	$\frac{35}{4}$	0	1	
8	0	0	1	0	21	0	14	0	1

Table 1. Values of $T_0(n, k)$ for $0 \leq k \leq n \leq 8$.

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	2	1							
2	4	4	1						
3	8	$\frac{49}{4}$	6	1					
4	16	34	25	8	1				
5	32	$\frac{1441}{16}$	90	$\frac{85}{2}$	10	1			
6	64	$\frac{931}{4}$	301	190	65	12	1		
7	128	$\frac{37969}{64}$	966	$\frac{12411}{16}$	350	$\frac{371}{4}$	14	1	
8	256	$\frac{6001}{4}$	3025	3003	1701	588	126	16	1

Table 2. Values of $T_2(n, k)$ for $0 \leq k \leq n \leq 8$.

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	2	1							
2	4	4	1						
3	8	$\frac{49}{4}$	6	1					
4	16	34	25	8	1				
5	32	$\frac{1441}{16}$	90	$\frac{85}{2}$	10	1			
6	64	$\frac{931}{4}$	301	190	65	12	1		
7	128	$\frac{37969}{64}$	966	$\frac{12411}{16}$	350	$\frac{371}{4}$	14	1	
8	256	$\frac{6001}{4}$	3025	3003	1701	588	126	16	1

Table 3. Values of $T_2(n, k)$ for $0 \leq k \leq n \leq 8$.

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	3	1							
2	9	6	1						
3	27	$\frac{109}{4}$	9	1					
4	81	111	55	12	1				
5	243	$\frac{6841}{16}$	285	$\frac{185}{2}$	15	1			
6	729	$\frac{12753}{8}$	1351	585	140	18	1		
7	2187	$\frac{372709}{64}$	6069	$\frac{53011}{16}$	1050	$\frac{791}{4}$	21	1	
8	6561	$\frac{167943}{8}$	26335	$\frac{35049}{2}$	6951	1722	266	24	1

Table 4. Values of $T_3(n, k)$ for $0 \leq k \leq n \leq 8$.

8. DOUBLE AND TRIPLE SUMS OF POWERS IN TERMS OF CENTRAL DIFFERENCES

Proposition 8.1 (Double sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^2 n^m = \frac{1}{2} \sum_{k=0}^m \left[+ \binom{n-t+\frac{k}{2}+2}{k+2} - \binom{2-t+\frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+1} \Sigma^1 n^0 \right. \\ \left. + \binom{n-t+\frac{k}{2}+1}{k+2} - \binom{1-t+\frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{0-t+\frac{k}{2}}{k+1} \Sigma^1 n^0 \right] \delta^k t^m \end{aligned}$$

Proposition 8.2 (Triple sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^3 n^m = \frac{1}{2} \sum_{k=0}^m \left[\right. \\ + \binom{n-t+\frac{k}{2}+3}{k+3} - \binom{3-t+\frac{k}{2}}{k+3} \Sigma^0 n^0 - \binom{2-t+\frac{k}{2}}{k+2} \Sigma^1 n^0 - \binom{1-t+\frac{k}{2}}{k+1} \Sigma^2 n^0 \\ \left. + \binom{n-t+\frac{k}{2}+2}{k+3} - \binom{2-t+\frac{k}{2}}{k+3} \Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+2} \Sigma^1 n^0 - \binom{0-t+\frac{k}{2}}{k+1} \Sigma^2 n^0 \right] \delta^k t^m \end{aligned}$$

Example 8.3 (Double sum of squares in central differences).

$$\begin{aligned} t=0: \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(1+n)^2(2+n)}{12}, \\ t=1: \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-1+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n^2(-1+n^2)}{12}, \\ t=2: \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-4-3n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(10+5n-4n^2+n^3)}{12}, \\ t=3: \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-7-6n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(32+23n-8n^2+n^3)}{12}, \\ t=4: \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-10-9n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(66+53n-12n^2+n^3)}{12}. \end{aligned}$$

Example 8.4 (Triple sum of squares in central differences).

$$\begin{aligned} t=0: \quad \Sigma^3 n^2 &= \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(6+11n+6n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(18+45n+40n^2+15n^3+2n^4)}{120}, \\ t=1: \quad \Sigma^3 n^2 &= \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-2-n+2n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(-2-5n+5n^3+2n^4)}{120}, \\ t=2: \quad \Sigma^3 n^2 &= \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-10-13n-2n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(58+65n-5n^3+2n^4)}{120}, \\ t=3: \quad \Sigma^3 n^2 &= \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-18-25n-6n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(198+255n+40n^2-15n^3+2n^4)}{120}, \\ t=4: \quad \Sigma^3 n^2 &= \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-26-37n-10n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(418+565n+120n^2-25n^3+2n^4)}{120}. \end{aligned}$$

9. MULTIFOLD SUMS OF POWERS IN TERMS OF CENTRAL DIFFERENCES

Proposition 9.1 (Multifold sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} \delta^k t^m \end{aligned}$$

Example 9.2 (Examples of multifold binomial sums).

$$\begin{aligned} t = 0 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n+\frac{k}{2}+r}{k+r} + \binom{n+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s+\frac{k}{2}}{k+r-s} + \binom{r-1-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 0^m, \\ t = 1 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-1+\frac{k}{2}+r}{k+r} + \binom{n-1+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-1+\frac{k}{2}}{k+r-s} + \binom{r-2-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 1^m, \\ t = 2 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-2+\frac{k}{2}+r}{k+r} + \binom{n-2+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-2+\frac{k}{2}}{k+r-s} + \binom{r-3-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 2^m. \end{aligned}$$

10. MULTIFOLD SUMS OF POWERS IN GENERALIZED CENTRAL FACTORIAL NUMBERS

Proposition 10.1 (Multifold central factorial sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} k! T_t(m, k) \end{aligned}$$

t	$T_t(2m, 2k)$	OEIS
0	1, 0, 1, 0, 1, 1, 0, 1, 5, 1, 0, 1, 21, 14, 1, 0, 1, 85, 147, 30, 1, $\dots$	A269945
1	1, 1, 1, 1, 7, 1, 1, 31, 20, 1, 1, 127, 231, 42, 1, 1, 511, $\dots$	A394692
2	1, 4, 1, 16, 25, 1, 64, 301, 65, 1, 256, 3025, 1701, 126, 1, $\dots$	A395456
3	1, 9, 1, 81, 55, 1, 729, 1351, 140, 1, 6561, 26335, 6951, 266, 1, $\dots$	A395457

Example 10.2 (Sums of squares in central factorial numbers).

$$t = 0 : \quad \Sigma^1 n^2 = n \cdot 0 + \frac{1}{2}n(1+n) \cdot 0 + \frac{1}{6}n(1+3n+2n^2) \cdot 1,$$

$$t = 1 : \quad \Sigma^1 n^2 = n \cdot 1 + \frac{1}{2}n(n-1) \cdot 2 + \frac{1}{6}n(1-3n+2n^2) \cdot 1,$$

$$t = 2 : \quad \Sigma^1 n^2 = n \cdot 4 + \frac{1}{2}n(n-3) \cdot 4 + \frac{1}{6}n(13-9n+2n^2) \cdot 1,$$

$$t = 3 : \quad \Sigma^1 n^2 = n \cdot 9 + \frac{1}{2}n(n-5) \cdot 6 + \frac{1}{6}n(37-15n+2n^2) \cdot 1,$$

$$t = 4 : \quad \Sigma^1 n^2 = n \cdot 16 + \frac{1}{2}n(n-7) \cdot 8 + \frac{1}{6}n(73-21n+2n^2) \cdot 1,$$

$$t = 5 : \quad \Sigma^1 n^2 = n \cdot 25 + \frac{1}{2}n(n-9) \cdot 10 + \frac{1}{6}n(121-27n+2n^2) \cdot 1.$$

Example 10.3 (Sums of cubes in central factorial numbers).

$$\begin{aligned}
t = 0 : \quad \Sigma^1 n^4 &= n \cdot 0 + \frac{1}{2}n(1+n) \cdot 0 + \frac{1}{6}n(1+3n+2n^2) \cdot 1 \\
&\quad + \frac{1}{8}n(-1+n+4n^2+2n^3) \cdot 0 + \frac{1}{10}n(-2-5n+5n^3+2n^4) \cdot 1, \\
t = 1 : \quad \Sigma^1 n^4 &= n \cdot 1 + \frac{1}{2}n(n-1) \cdot 5 + \frac{1}{6}n(1-3n+2n^2) \cdot 7 \\
&\quad + \frac{1}{8}n(1+n-4n^2+2n^3) \cdot 4 + \frac{1}{10}n(-2+5n-5n^3+2n^4) \cdot 1, \\
t = 2 : \quad \Sigma^1 n^4 &= n \cdot 16 + \frac{1}{2}n(n-3) \cdot 34 + \frac{1}{6}n(13-9n+2n^2) \cdot 25 \\
&\quad + \frac{1}{8}n(-21+25n-12n^2+2n^3) \cdot 8 + \frac{1}{10}n(18-45n+40n^2-15n^3+2n^4) \cdot 1, \\
t = 3 : \quad \Sigma^1 n^4 &= n \cdot 81 + \frac{1}{2}n(n-5) \cdot 111 + \frac{1}{6}n(37-15n+2n^2) \cdot 55 \\
&\quad + \frac{1}{8}n(-115+73n-20n^2+2n^3) \cdot 12 + \frac{1}{10}n(298-275n+120n^2-25n^3+2n^4) \cdot 1, \\
t = 4 : \quad \Sigma^1 n^4 &= n \cdot 256 + \frac{1}{2}n(n-7) \cdot 260 + \frac{1}{6}n(73-21n+2n^2) \cdot 97 \\
&\quad + \frac{1}{8}n(-329+145n-28n^2+2n^3) \cdot 16 + \frac{1}{10}n(1318-805n+240n^2-35n^3+2n^4) \cdot 1, \\
t = 5 : \quad \Sigma^1 n^4 &= n \cdot 625 + \frac{1}{2}n(n-9) \cdot 505 + \frac{1}{6}n(121-27n+2n^2) \cdot 151 \\
&\quad + \frac{1}{8}n(-711+241n-36n^2+2n^3) \cdot 20 + \frac{1}{10}n(3798-1755n+400n^2-45n^3+2n^4) \cdot 1.
\end{aligned}$$

11. MULTIFOLD SUMS OF POWERS IN BINOMIAL FORM

Lemma 11.1 (Multifold sum of zero powers). *For integers $r \geq 0$ and $n \geq 1$*

$$\Sigma^r n^0 = \binom{r+n-1}{r}$$

Proof. (1) Let $r = 0$, then we have $\Sigma^0 n^0 = 1$, by definition (1.1).

(2) Let $r = 1$, then we have $\Sigma^1 n^0 = \sum_{k=1}^n 1 = \binom{n}{1}$.

(3) Let $r = 2$, then we have $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \binom{n+1}{2}$.

(4) Let $r = 3$, then we have $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$.

(5) Similarly, for arbitrary r , by hockey stick identity $\sum_{k=1}^n \binom{k}{r} = \binom{n+1}{r+1}$, thus, the claim follows $\Sigma^r n^0 = \sum_{k=1}^n \Sigma^{r-1} k^0 = \sum_{k=1}^n \binom{k+r-2}{r-1} = \binom{r+n-1}{r}$.

□

Proposition 11.2 (Multifold binomial sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta_{t^m}^{k_t} \end{aligned}$$

12. RELATED WORKS

13. FRAMEWORK FOR SUMS OF POWERS

14. CONCLUSIONS

15. MATHEMATICA PROGRAMS

Use this [GitHub](#) repository to validate the results using Mathematica programs.

Mathematica Function	Validates / Prints
<code>MultifoldSumOfPowersRecurrence[r, n, m]</code>	Computes $\Sigma^r n^m$, (1.1)
<code>CentralFactorial[n, k]</code>	Computes $n^{[k]}$, (1.2)
<code>ValidateHalvedCentralBinomials[20]</code>	Validates Lem. (4.2)
<code>ValidateNewtonsFormulaForPowers[10]</code>	Validates Prop. (4.3)
<code>ValidateHalvedCentralFactorialPowers[5]</code>	Validates Prop. (4.4)
<code>ValidateCenteredHockeyStickIdentity[5]</code>	Validates Lem. (4.6)
<code>ValidateOrdinarySumsOfPowersDecomposition[10]</code>	Validates Prop. (5.2)
<code>ValidateDoubleSumsOfPowersDecomposition[5]</code>	Validates Prop. (8.1)
<code>ValidateTripleSumsOfPowersDecomposition[5]</code>	Validates Prop. (8.2)
<code>ValidateMultifoldSumsOfPowers[3]</code>	Validates Prop. (9.1)